

OPEN INNOVATION · ARTIFICIAL INTELLIGENCE



LogiSolve

*Autonomous Multi-Constraint Production Planning
& Transportation Optimization Agent*

Team Red&Yellow

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SECTION 1

Project Overview

LogiSolve is an email-driven AI agent that turns a plain-English planning request into a fully optimized, cost-minimizing schedule — in seconds. It now solves two real supply-chain problems through one system: production scheduling and delivery route planning.

Why judges should care

Not a chatbot demo — a working pipeline where AI only reads language, and a mathematical solver guarantees the numbers are actually correct and optimal.



One inbox, zero UI

Every request — production or delivery — is a normal email. No dashboard, no login, no training needed.



Two capabilities, one engine

Multi-product production planning and multi-stop transportation routing, both solved with the same architecture.



AI translates, solver calculates

The language model never does arithmetic — a proven optimization solver does, so answers are always correct.

SECTION 2

Problem Statement

Mid-sized manufacturers lose money at three separate points in their planning cycle — long before a schedule is ever executed.



Unstructured Inputs

Planning requests arrive as free-form, chaotic emails — no standard format, no structured handoff. Every cycle starts with a human reading and interpreting raw text.



LLM Computation Failures

General-purpose AI makes arithmetic errors, violates capacity limits, and hallucinates numbers when asked to solve optimization problems directly.



Cost Inefficiencies

Manual, spreadsheet-based scheduling leans on expensive overtime and poor inventory decisions — sub-optimal choices that quietly erode margin every cycle.

SECTION 3

Proposed Solution

A hybrid neuro-symbolic pipeline: the AI reads and classifies the request, a deterministic solver does every calculation.



The solver step automatically routes to the right optimization model:

Production Branch

Multi-product capacitated lot-sizing (CLSP) — shared machine capacity, setup, holding & overtime costs optimized together.

Transportation Branch

Multi-stop vehicle routing — capacity-aware vehicle assignment, real road distances, baseline vs. optimized cost comparison.

SECTION 4

Live Prototype & Demo



Gmail trigger fires on a new request



AI Agent classifies: production or transportation



Production → multi-product MILP solver



Transportation → route optimizer



Formatted reply sent back to the thread

Actual working n8n orchestration workflow — screenshot from the live build

SECTION 5

Technologies Used



n8n

Event listening, thread isolation, orchestration



LLM API

Semantic parsing of free-form email text



Python FastAPI

High-performance REST layer for the solver



PuLP + CBC

Deterministic MILP optimization engine



OpenRouteService

Real road distances & travel times



Gmail API

Secure send / receive / thread tracking

Key Features



Multi-Product Planning

Several products share one machine's capacity per period — not solved one at a time.



Cost-Balanced Scheduling

Setup, holding, and overtime costs optimized together, every cycle.



Transportation Optimization

Multi-stop delivery routing with capacity-aware vehicle assignment.



Real Road Distances

Live routing data, not straight-line estimates — realistic cost numbers.



Automatic Intent Routing

One inbox detects production vs. transportation requests on its own.



Zero UI Requirement

If a planner can send an email, they can use LogiSolve.

Results & Economic Impact

<10 sec

to a formatted, ready-to-use plan
(vs. hours of manual spreadsheet work)

100%

mathematically optimal schedules —
zero hallucinated production numbers

2-in-1

platform: production planning +
transportation, one architecture

Where the money is saved

- Planner time: hours of manual scheduling collapse into a single email round-trip.
- Production cost: overtime and inventory holding are minimized automatically, every cycle — not just when someone remembers to optimize.
- Transportation cost: fewer vehicles, shorter routes — every plan compares baseline vs. optimized cost so the saving is visible, not assumed.
- Adoption cost: zero new software to buy, install, or train on — the interface is the inbox everyone already uses.

SECTION 8

Future Scope

Transportation route optimization — once "future scope" — is already live. Here's what's next.



Dynamic Safety Stock

Rolling-horizon demand forecasting to automatically handle demand uncertainty and lead-time volatility.



Carbon Footprint Tracking

CO₂ emission metrics per batch and per route, so sustainability caps can sit alongside cost limits.



Deeper Multi-Echelon Reach

Extending shared-capacity planning and routing across multiple factories and warehouses, not just one site.



Enterprise Integration

Direct connectors into existing ERP / inventory systems, so LogiSolve can sit inside current workflows.



Thank You

Questions & Discussion

LogiSolve

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github.com/jerrilpritamdsa/SCM-Agentic-AI